Honey Sahu

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Skills

- **Languages**: C, C++, Python, Java, SQL
- **Libraries**: NumPy, Pandas, Scikit-learn, NLTK, Matplotlib
- **Tools/Platforms**: Git, GitHub, Jira, Jupyter Notebook, VS Code
- **Soft Skills**: Problem Solving, Team Player, Time Management, Adaptability

Projects

- **Sentiment Analysis Chatbot using ML**: GitHub April 2025
  - Created an ML-based chatbot to detect user sentiment and provide supportive mental health responses.
  - Handled a 1.6M-tweet dataset with text cleaning and TF-IDF feature extraction.
  - Evaluated several models and chose Logistic Regression for its 80% accuracy.
  - Integrated sentiment-driven response logic and launched via an intuitive Streamlit interface.
  - Produced a reliable chatbot offering precise sentiment detection and empathetic real-time interactions.

  Tech: Python, Pandas, Scikit-learn, NLTK, Streamlit, Jupyter Notebook

- **Spam SMS Classifier using ML**: GitHub July 2025
  - Developed a machine learning-based system to classify SMS messages as spam or ham using natural language processing techniques.
  - Preprocessed text data through tokenization, stopword removal, and TF-IDF vectorization to enhance model performance.
  - Implemented and trained a Random Forest classifier, achieving high accuracy and robust generalization on unseen data.
  - Engineered a complete pipeline covering data cleaning, feature extraction, model training, and evaluation with performance metrics.
  - Delivered an efficient spam detection solution capable of improving message filtering and reducing unwanted communication.

  Tech: Python, Pandas, NumPy, Scikit-learn, NLTK, Jupyter Notebook

- **Noise Pollution Zone Clustering for City Planning using ML**: GitHub November 2025
  - Performed unsupervised analysis on long-term noise monitoring data from 70 stations across 7 Indian cities.
  - Prepared and aggregated CPCB datasets with feature engineering on noise levels, exceedance values, and violation rates.
  - Implemented K-Means clustering with Elbow and Silhouette methods, identifying four distinct noise pollution zones.
  - Analyzed discrepancies between official zone labels and real-world noise behavior, highlighting severe violations in Silence and Residential areas.
  - Visualized cluster separation using PCA to support urban planning insights and targeted intervention strategies.

  Tech: Python, Pandas, NumPy, NLTK, Scikit-learn, Matplotlib

Certificates

- Communication in the 21st Century Workplace - University of California, Irvine - Link March 2026
- Master Generative AI & Generative AI tools - Infosys - Link August 2025
- Build Generative AI Apps and Solutions with No-Code Tools - Infosys - Link August 2025
- Computational Theory: Language Principle & Finite Automata Theory - Infosys - Link August 2025
- Fundamentals of Data Structures using C++ - LPU - Link July 2025
- Java Programming - IamNeo - Link May 2025
- Learn C++ Programming - Beginner to Advanced - Deep Dive in C++ - Udemy - Link November 2024

Achievements

- Solved **130+ Data Structures and Algorithms problems** on LeetCode. March 2026

Education

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 8.13*** *Since August 2023*
- **Kaliram Chandrakar Public School** Kurud, Chhattisgarh
  Intermediate; ***Percentage: 81*** *April 2021 - March 2022*
- **Kaliram Chandrakar Public School** Kurud, Chhattisgarh
  Matriculation; ***Percentage: 93*** *April 2019 - March 2020*